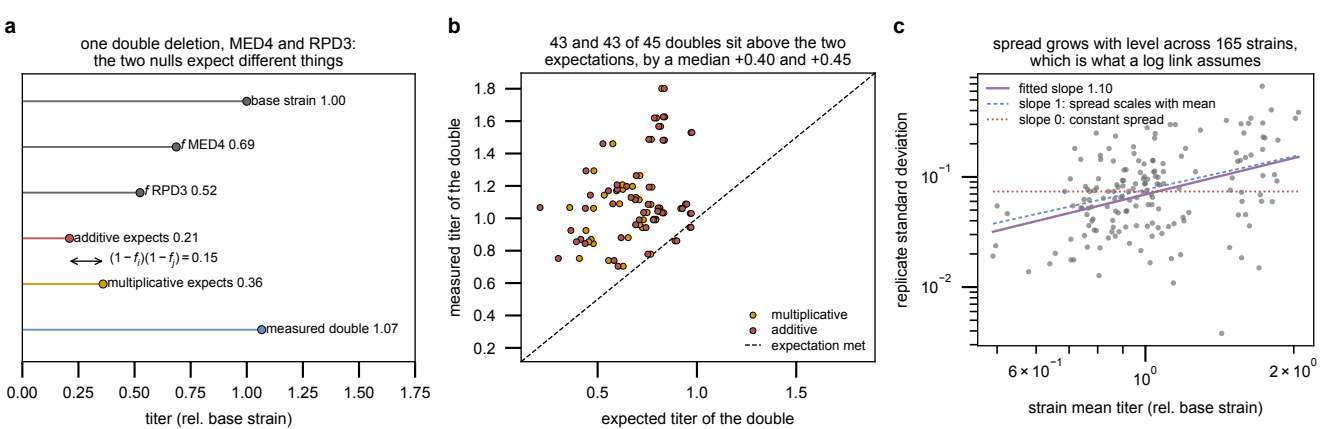
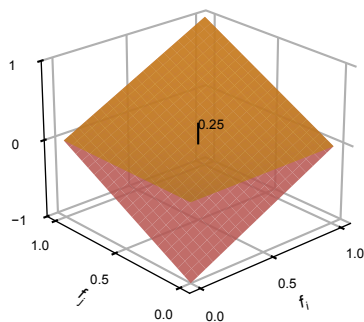
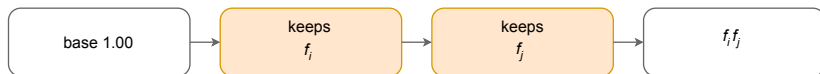




d expected f_{ij} over the single-deletion square; both log-scale models share the multiplicative surface

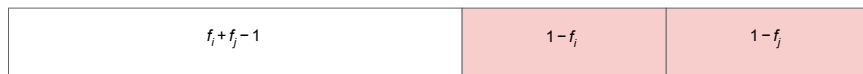


e effects are FRACTIONS: two steps in series



a fraction of a fraction: the second step only ever acts on what the first let through

effects are AMOUNTS: two draws on one pool



each loss is taken from the whole pool, so the part the first deletion already removed is counted a second time

the two nulls differ by exactly $(1 - f_i)(1 - f_j)$, the loss the second deletion would take again

f

model	expects, for a double	residual measured on	fit to, and what that assumes	the reading that makes it the natural null
multiplicative	$f_i f_j$	linear titer	strain means; one standard error per strain, propagated by the delta method	each deletion keeps a fixed FRACTION of what reaches it, so independent steps of one flux compose; the null of the growth screens
additive	$f_i + f_j - 1$	linear titer	strain means; one standard error per strain, propagated by the delta method	each deletion removes a fixed AMOUNT from a shared pool, closer to how a titer in mg/L is built up
GLM log-link	$\exp(\alpha_i + \alpha_j)$	log titer	replicate titers; Gamma family, spread proportional to the mean (panel c)	the readout is positive and noisier where it is larger, and every strain's replicates are used rather than its mean
log-OLS	$\exp(\alpha_i + \alpha_j)$	log titer	replicate log ratios to the base strain; constant spread on the log scale	the question is whether a deletion's FOLD effect carries into a new background, which is a log-scale question